

Soft Identity Constraints for LLM-Written Knowledge Graphs

A vision for reliable graph memory under incremental ingestion

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1 Situation

Knowledge graphs are becoming a memory substrate for AI agents. Systems such as Graphiti, HippoRAG, GraphRAG, and Mem0 convert conversations and documents into entities and relations, then retrieve context through graph search and random walks. Unlike a conventional database, however, these graphs are written by a probabilistic model operating on partial context. The engine still assumes stable entity identity, valid edges, consistent schemas, and ordered updates, although no human curator now guarantees those invariants [3, 1, 2, 4, 10]. This mismatch produces structural, temporal, and semantic faults: one entity is split across aliases; distinct entities are merged into a generic hub; relations drift in name and granularity; stale facts remain active; and provenance is flattened. Evidence from research and production systems points especially to entity resolution. Reported duplication reaches 27–36% in one document study, resolution degrades as context grows, and production pipelines combine exact matching, embedding retrieval, and LLM adjudication without claiming the problem is solved [5, 6, 7, 8, 9]. Incremental ingestion is the critical setting for agent memory: each episode arrives without the global view a human curator would use to enforce **UNIQUE**.

2 Vision: Integrity Mechanisms for Probabilistic Writers

The long-term goal is an “immune system” for machine-written graphs: integrity mechanisms enforced by the storage and retrieval engine rather than delegated to an extraction prompt. Each violated invariant should have a measurable fault, a write-time guard, a reversible repair, and an end-to-end benchmark. Identity is the first work package because both under-resolution and over-resolution alter the paths through which an agent assembles evidence.

The central design principle is that uncertain identity should not be forced into a binary merge decision. On insertion, the engine should retain source nodes and provenance, retrieve possible aliases, and write provisional **same_as** edges with calibrated confidence. At read time, retrieval may traverse those edges in proportion to confidence. A correct provisional link reconnects scattered facts; an incorrect link has bounded influence and can be revoked. This is safer than an irreversible hard merge and more useful than leaving likely aliases disconnected. Existing systems already use multi-stage resolution [3, 5, 9]. This motivates a separate question: should uncertainty be exposed to retrieval rather than collapsed into a single merge decision?

For a node v with identity confidence c_v , a simple retrieval operator reserves a bounded share of its outgoing probability for identity transitions:

$$P(v, \cdot) = (1 - \lambda c_v) P_{\text{graph}}(v, \cdot) + \lambda c_v P_{\text{identity}}(v, \cdot),$$

where λ caps the effect of identity uncertainty. The broader research vision extends the same pattern to temporal validity, schema equivalence, provenance, and concurrent writers.

3 Preliminary Evidence

Our strongest preliminary result shows one identity failure and a partial repair. We used MuSiQue, a public collection of questions whose answers require combining facts from two or more source paragraphs, meaning short sections of the original documents [11]. The dataset identifies which paragraphs contain the required facts. Claude and Codex independently converted the same 1,048 paragraphs into two graphs. We evaluated macro-averaged supporting-passage Recall@5: the fraction of gold supporting paragraphs present in the top five retrieved results, averaged over questions.

We restricted the test to questions with two required paragraphs that shared an entity. In the intact graph, one entity node was connected to facts from both paragraphs. We then simulated an identity error by splitting it into two disconnected nodes: the first kept only the edges extracted from paragraph A, and the second kept only those from paragraph B. We began retrieval at the first node and measured whether paragraph B appeared in the top five results. Finally, we connected the two nodes with a known-correct **same_as** edge and repeated retrieval. The table reports the score before the split, after the split, and after reconnection. Its final column is the fraction of the lost score restored by the link:

Extractor	Intact graph	Split graph	With link	Loss recovered
Claude Opus 5	75.0%	15.0%	64.2%	81.9%
Codex GPT-5.6-sol	77.3%	10.0%	71.8%	91.9%

The changes were consistent across questions: repeatedly resampling them produced 95% uncertainty ranges that always showed a loss after splitting and a gain after linking. This demonstrates the mechanism under controlled conditions, not yet a common production failure. We created the split and supplied the correct link ourselves. We have not measured how often real ingestion creates duplicate nodes or whether an automatic resolver finds the right link. The focused test was also designed after a broader experiment found no effect, so it needs an independent replication fixed in advance.

We separately tested “supernodes,” nodes with unusually many edges. Some graph retrievers rank nodes by repeatedly simulating a walk along edges [1, 2]. A supernode may then appear important for many unrelated questions. In one 40-part conversation, the same highly connected people ranked near the top for many different queries. Penalizing nodes with many and varied edges reduced this concentration, but did not reliably improve retrieval of the correct source paragraphs. The supposed bad supernodes were often legitimate people or places, and a simple degree penalty worked as well as the proposed relation-aware score. Related work also finds that extraction errors do not change graph statistics uniformly, and that graph retrieval benefits depend strongly on the task and construction pipeline [8, 10].

This negative result may reflect scale: the study used one conversation and 60 multi-source questions. A larger and more diverse experiment could reveal a smaller effect. Scale is not assumed to be the explanation, however. The proposed score may be wrong, or direct text similarity may bypass the graph error. A larger study with its analysis fixed in advance should distinguish these possibilities.

4 Research Programme

Aim 1: establish the natural fault. Compare the same extractor and corpus under (A) batch extraction with global context, (B) blind incremental extraction, and (C) incremental extraction with resolution. Repeat extraction to measure sampling instability. Manually adjudicate mention-to-entity clusters on a fixed sample. Primary outcomes are the duplicate-node rate, how an entity’s facts are divided among its duplicates, and the percentage of required sources found in the first five results. Questions answerable from one source are the control. The prediction is conditional: fragmentation should hurt when it breaks a necessary evidence path, not uniformly across all questions [5, 6, 9].

Aim 2: build reversible resolution. Implement a cost-aware cascade: normalized exact matching, semantic similarity to propose candidates, contextual scoring, and LLM adjudication only for ambiguous cases. Preserve original nodes and attach calibrated confidence to provisional identity edges. Compare no resolution, hard merge, soft identity flow, and known-correct links under identical extraction and retrieval budgets.

Aim 3: test safety and generality. Inject incorrect identity links to measure false bridging and compare graceful degradation with destructive hard merging. Run on LoCoMo and MuSiQue, then confirm on an untouched multi-document or long-memory benchmark. Replicate across at least two extractor families. Tune on development data and evaluate once on questions set aside before tuning; report uncertainty across questions.

Success gates. Natural incremental ingestion must produce more fragmentation than batch ingestion, with retrieval loss concentrated on bridge-dependent questions. The practical resolver should recover at least half of the improvement achieved by known-correct links, create fewer false bridges than hard merging, and average fewer than one LLM adjudication call per episode. If natural harm is not observed, the result will be reported as a robustness mechanism rather than evidence of a common production fault.

5 Expected Contribution and Scope

The project contributes: (1) a controlled characterization of identity failure under incremental graph construction; (2) an evaluation focused on questions that require an identity connection; and (3) a reversible, confidence-weighted retrieval rule for identity links. The immediate deliverable is deliberately narrow: prove the natural fault, then test whether soft resolution repairs it safely. Temporal consistency, schema drift, provenance policy, and multi-agent concurrency remain future work within the larger integrity-mechanism vision.

References

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